

CLOSING THE SYNTHETIC-TO-REAL GAP IN SCORE FOLLOWING ON SHEET MUSIC IMAGES

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ABSTRACT

Score following on sheet music images tracks a live performance directly on a page of printed music, with no symbolic transcription of the score. Such systems work on synthesized audio and fail on real recordings, and closing that gap is the field’s central open problem: the strongest published system tracks 95.5% of onsets within 0.5 s on synthetic audio and 88.3% on real piano recordings, and earlier systems fall much further. Every remedy proposed so far has treated the gap as a perception failure. We show it is not. On real recordings the perception model already proposes a correct position at every scored onset, and an oracle restricted to causal, mono-tone choices among its proposals reaches 99.6% where the deployed system reaches 80.0%. We present CANDOR, which follows the score by deciding sequentially among the positions the page affords, combining a transition model over displacements with a set-based scorer that judges each position against its competitors. CANDOR tracks 94.4% of onsets within 0.5 s of real piano recordings against 88.3% for the previous best, and holds that margin across three independently trained perception models. Analysis of the remaining error shows that the image representation is irrelevant while the temporal history is decisive, that accuracy on synthesized audio barely predicts accuracy on real audio, and that the forward-motion assumption fails at repeats.

Index Terms— Score following, sheet music, audio-to-score alignment, real-time, music information retrieval

1. INTRODUCTION

Score following is the real-time alignment of a live performance to its score, underlying automatic page turning [1] and computer accompaniment [2]. Classical systems require the score in symbolic form and track the performance over a discrete set of score states [3, 4, 5], but most music exists only as printed or scanned paper and optical music recognition is still too error-prone to supply that form automatically [6]. Sheet-image score following removes the requirement: the system reads a page of pixels and a stream of audio and reports where on the page the musician is [7, 8, 9, 10, 11, 12].

The gap. Training such a system needs performances aligned to notehead coordinates on the page, an annotation

that is expensive enough that essentially all of it is produced by synthesis: a score is rendered to MIDI, the MIDI to audio, and the alignment comes for free. Systems trained this way work on synthesized audio and degrade sharply on a real instrument in a real room. The published figures are stark. Between synthetic audio and real piano recordings of the same pieces, CUNet [9] falls from 85.5% of onsets tracked within 0.5 s to 22.4%, CYOLO-SB [11] from 89.3% to 79.9%, and the strongest published system CODA [12] from 95.5% to 88.3%. Every remedy proposed has been a perception remedy: reverberant augmentation [10], extra proprietary recordings [11], or a new end-to-end architecture [12].

Perception is not the bottleneck. We tested that premise before accepting it. A conditional detector for this task does not produce a position directly; it scores thousands of candidate noteheads on the page and reports the most confident one. Keeping the 256 most confident, we find that on real room-microphone recordings a candidate within the 0.5 s tolerance is present at *every one* of 4,149 scored onsets, and that an oracle which must commit one step at a time and may never move backwards tracks 99.6% of them. The deployed rule tracks 80.0% (Fig. 1). Nineteen of the twenty points lost to real audio are still recoverable from what the model already saw. The room degrades the confidences, not the evidence.

This paper. We therefore treat score following on a sheet image as a sequential decision over the positions the page affords, rather than as the regression of a coordinate. CANDOR makes that decision with two parts: a transition model that scores a displacement against how much time has passed, and a set-based scorer that judges each candidate position against its competitors in the same instant, trained listwise on synthesized audio alone. On real piano recordings CANDOR tracks 94.4% of onsets within 0.5 s, against 88.3% for the best published system, and it reaches that figure without layout annotations at test time.

Our contributions are: (i) evidence, from candidate-set oracles, that this gap is a decision failure and not a perception failure; (ii) CANDOR, which acts on it and sets a new best on real recordings by a margin holding across three independently trained perception models; (iii) an analysis of what remains, covering the irrelevance of the image representation, the unreliability of synthesized audio for model selection, and the failure at discontinuities.

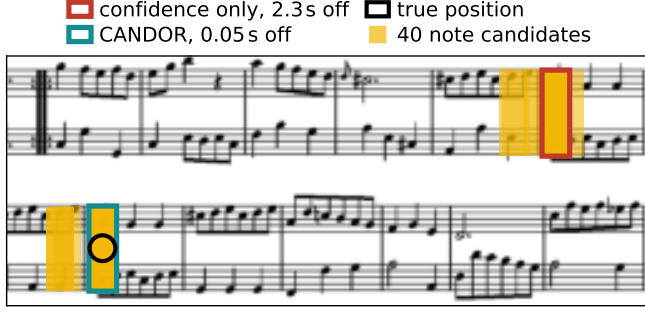


Fig. 1. One instant of a real room-microphone recording. Yellow: 40 of the candidate positions the detector scores. The most confident (red) lies on the wrong staff, 2.3 s from the truth; CANDOR (teal) takes a less confident candidate on the true note (ring). The correct position was available and was not chosen, which is the characteristic failure on real audio.

2. RELATED WORK

Sheet-image score following. Three approaches have been tried. *Dense localizers* treat the page as a canvas: MM-Loc [7] and CUNet [9] emit an audio-conditioned heatmap and read a coordinate off its centre of mass. *Agents* treat it as control: a reinforcement-learning policy [8] moves a reading head, rewarded for staying aligned. *Conditional detectors* treat it as a query: CYOLO [10] encodes the recent audio into a vector and runs object detection, so the audio asks which part of the page it sounds like, and CYOLO-SB [11] adds heads for the enclosing system and bar. The concurrent CODA [12] cascades system, bar and note selection over the layout regions of an annotated score, which it requires at test time.

A common perception module. Beneath these differences the field has settled on one visual front end. In each case a convolutional network reads the page and its feature maps are modulated by feature-wise linear modulation [13], $f_{\text{FiLM}}(x) = s(z) \odot x + t(z)$, from a vector z encoding the recent audio; CYOLO applies FiLM in its downscale and up-scale blocks and CODA at the deeper stages of its backbone. Our diagnosis concerns that module, and our experiments use three separately trained instances of it, so the conclusions are about the family rather than about one checkpoint.

Decision over candidates. Choosing among scored proposals under a structural constraint is familiar from detection [14] and from symbolic score following, where the score supplies the state set. Here the state set is neither given nor hand-built: perception emits it and then discards it.

3. CANDOR

3.1. Candidate positions

A performance is processed in instants $t = 1, \dots, T$. At each instant the perception model reads the audio so far and

the current page and scores note boxes $b_i = (x_i, y_i, w_i, h_i)$ with confidence o_i ; we keep the $K = 256$ most confident, $\mathcal{C}_t$. Staves are concatenated along one horizontal axis, the unrolled coordinate u_i , so positions across a line break are adjacent. The released detectors score 8,112 boxes per instant on a 52×52 grid and report

$$\hat{i}_t = \arg \max_{i \in \mathcal{C}_t} o_i. \quad (1)$$

CANDOR replaces Eq. 1 and reads only $\mathcal{C}_t$ and its own previous choice $\hat{u}_{t-1}$.

3.2. Transition model

Successive decisions must describe a plausible trajectory. Between instants Δ_t frames apart the performer advances by a roughly predictable amount, so the displacement $d_i = u_i - \hat{u}_{t-1}$ is scored by a floored Gaussian whose mean grows with elapsed time,

$$\log \pi_i = \max \left(-\frac{(d_i - \mu \rho_t)^2}{2\sigma^2}, J \right), \quad (2)$$

$\rho_t = \text{clip}(\Delta_t/\Delta_0, 0.2, 8)$ with $\Delta_0 = 5$ frames, the mean inter-onset gap in training. The floor J keeps a jump possible at bounded cost. Combined with confidence, $h_i = \log o_i + \log \pi_i$ already recovers a large part of the gap on its own.

3.3. Set-based scorer

The transition model treats candidates independently given the history. What it cannot express is comparative: whether *this* box, among those on offer in this instant, looks like a real notehead rather than a spurious response. Each candidate is described by $\phi_i \in \mathbb{R}^{33}$, holding its confidence relative to the instant’s best, box geometry, motion relative to the history, relation to any coarse regions the model emits, and a neighbourhood group (how many candidates fall within 25 and 100 px, spacing to the nearest on either side, rank in reading order, size against the instant’s median). We also read f_i , the 128-dimensional FiLM-conditioned feature from which the detection head scored the box, so the decision sees the representation perception used. A permutation-invariant network [15] embeds the set and pools a context shared by all candidates:

$$e_i = g([\phi_i; \text{ReLU}(W f_i)]), \quad c = \left[\frac{1}{K} \sum_j e_j; \max_j e_j \right], \quad (3)$$

$$s_i = r([e_i; c]), \quad (4)$$

with g, r two-layer perceptrons (64, 32 and 64, 1 units) and $W \in \mathbb{R}^{64 \times 128}$. The maximum is taken per dimension, so every candidate is judged against its competitors rather than in isolation.

3.4. Training and inference

The decision is supervised listwise. With ε_i the error in frames of candidate i , the target is $q_i \propto \exp(-\varepsilon_i/\tau)$, $\tau = 3$, and

$$\mathcal{L} = - \sum_{i \in \mathcal{C}_t} q_i \log \frac{\exp s_i}{\sum_j \exp s_j}, \quad (5)$$

so the scorer learns to order a set rather than to regress a coordinate. History features come from previous decisions, perturbed on 30% of samples by Laplace noise of scale 30 px to expose training to the drift of sequence prediction [16]. At inference CANDOR decides

$$\hat{i}_t = \arg \max_{i \in \mathcal{C}_t} \beta s_i + (1 - \beta) h_i, \quad (6)$$

causally and greedily, one pass over 256 candidates per instant, in 1.25 ms on a single CPU thread, well inside the 50 ms real-time budget at 20 fps. Training the scorer on the residual $s_i - h_i$ and deciding $\arg \max_i (s_i + h_i)$ is Eq. 6 at $\beta = 0.5$ and eliminates all four scalar constants; that variant is reported below.

4. EXPERIMENTAL SETUP

Data. The decision model is trained on 353 pieces of MSMD [17], synthesized and convolved with measured room impulse responses [18], and validated on 19. The evaluation set is the real-performance benchmark of [8]: 16 pieces, 25 pages and 4,149 onsets played on a Yamaha AvantGrand N2 and captured with a room microphone. A held-out set of 80 further pieces is rendered with a real impulse response and noise from 12 to 0.5 dB SNR. Optimization uses AdamW ($3 \cdot 10^{-3}$, cosine, 40 epochs, batch 256).

Metric. The ratio of onsets whose predicted position maps to within 0.05 to 5 s of the true time, with 0.5 s as the primary figure and 95% intervals bootstrapped over pieces [19].

Constants. (μ, σ, J, β) are fitted leave-one-piece-out: for each evaluation piece they are chosen on the other 15, pieces weighted equally, so no piece influences its own decoding. Every CANDOR figure is cross-validated this way, and the residual variant needs no fit at all. Seeds move results by about a point, so all rows are seed means with deviations.

5. RESULTS

5.1. A new best on real recordings

Table 1 sets CANDOR against the published literature on the same recordings. It tracks 94.4% of onsets within 0.5 s, where the previous best was 88.3%, and leads at every threshold. The margin is not an artifact of one perception model: the decision model is retrained on whichever candidates each model emits, and the gain is 20.4, 10.7 and 7.9 points on three of them. On the weakest, whose training data is entirely public, CANDOR passes the point at which the strongest one begins. A

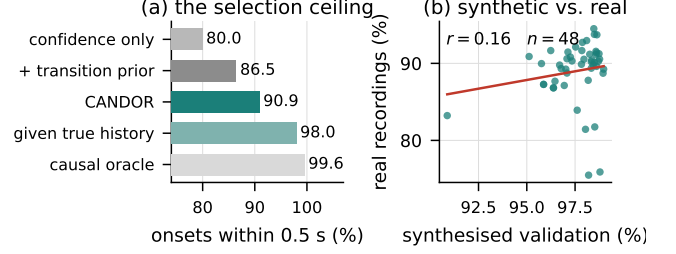


Fig. 2. (a) Every step from the deployed rule to the ceiling on the real recordings; CANDOR to *given true history* is drift, above it is ranking. (b) Each point is one of 48 trained models: accuracy on synthesised validation audio against accuracy on the real recordings.

decision model does not need good proposals, only proposals containing the answer.

The tightest thresholds qualify this: on CYOLO-SB candidates CANDOR reaches .690 and .714 at 0.05 and 0.1 s, below CODA’s .721 and .743, because a decision is only as precise as the boxes it chooses between. Deciding recovers which notehead, not where within it.

Table 2 separates the two parts. The transition model alone, with no learned parameters, accounts for roughly half the gain, and the residual variant, which fits no constant to any evaluation data, still reaches 88.7%.

5.2. What the decision depends on

Not the image. Giving the scorer the FiLM-conditioned backbone feature, DINOv2 patch features [20] on the same grid, or no image feature at all yields 90.6, 90.9 and 90.9, a spread well inside the 0.9-point seed deviation. On synthesized validation audio the same three differ by 3.4 points, so those features are learnable and useful there and worth nothing on real audio. Replacing the perception model’s audio encoder does not help either: a MERT tower [21] costs more than 40 points, and **Mamba pending**. The oracle predicted this.

The history. Removing the features computed against the previous decision costs 7.2 points, the largest single ablation here. Teacher forcing attributes about two thirds of the remaining error to drift: given the true previous position the same scorer reaches 98.0%, against the causal oracle’s 99.6% (Fig. 2a). What is hard is the sequence, not the image.

Robustness. Under a real impulse response and additive noise on 80 held-out pieces the scorer improves on the transition model at every level from 12 to 0.5 dB, by less as the SNR falls. On the 94 synthetic test pieces of the standard benchmark, where $\arg \max$ reaches .893, CANDOR reaches .955. Halving the candidate budget to $K = 128$ costs 0.5 points: the tail of the set carries real weight.

Synthesized audio does not predict real audio. Across 48 trained models, accuracy on the synthesized validation

Table 1. Real piano recordings: 16 pieces, 4,149 onsets, Yamaha AvantGrand N2 captured with a room microphone. Ratio of onsets tracked within each error threshold. [†] perception model trained with additional proprietary recordings, public checkpoint. *Layout*: system and bar annotations required at test time. Δ is the gain at 0.5 s over the perception model the row is built on. Published figures as reported in [12].

Method	Onset error threshold (s)					Layout	Δ
	≤ 0.05	≤ 0.1	≤ 0.5	≤ 1.0	≤ 5.0		
MM-Loc [7]	.364	.406	.585	.611	.735	no	
RL agent [8]	.185	.200	.476	.603	.901	no	
CUNet [9]	.113	.125	.224	.266	.443	no	
CYOLO [10]	.563	.581	.712	.749	.919	no	
CYOLO-SB [11]	.610	.630	.799	.832	.960	no	
CYOLO-SB+A [†] [11]	.682	.706	.865	.891	.981	no	
CODA [12]	.721	.743	.883	.914	.987	yes	
CANDOR on CYOLO	.726	.747	.916	.950	.991	no	+20.4
CANDOR on CYOLO-SB	.690	.714	.906	.942	.982	no	+10.7
CANDOR on CYOLO-SB+A [†]	.760	.785	.944	.969	.996	no	+7.9

Table 2. Components of CANDOR, cross-validated per piece on the real recordings at 0.5 s. *argmax* is Eq. 1 over the same candidates, *+transition* adds Eq. 2, *full* adds the scorer.

	<i>n</i>	<i>argmax</i>	<i>+transition</i>	<i>full</i>
<i>Perception model</i>				
CYOLO [10]	3	71.3	72.5	91.6 $\pm$ 1.4
CYOLO-SB [11]	6	80.0	86.5	90.6 $\pm$ 0.9
CYOLO-SB+A [†]	3	86.5	90.8	94.4 $\pm$ 0.9
<i>Decision model, on CYOLO-SB candidates</i>				
residual, no fitted constants	3	80.0	86.5	88.7 $\pm$ 0.9
without temporal history	3	80.0	86.5	83.7 $\pm$ 0.4
<i>Image features f_i read by the scorer</i>				
FiLM-conditioned backbone	6	80.0	86.5	90.6 $\pm$ 0.9
DINOv2 [20]	1	80.0	86.5	90.9
none	1	80.0	86.5	90.9
<i>Audio encoder replaced, perception model retrained</i>				
MERT [21]	3	33.4	31.9	46.2 $\pm$ 9.1
Mamba [22]	3	.	.	.

split correlates $r = 0.16$ with accuracy on the real recordings (Fig. 2b); the best-validation model scores 89.3 on real audio, below one reading no image features at all. A reverberant held-out set does better ($r = 0.80$) and still mispicks. Since the whole field selects models on synthesized audio, this is a caution beyond our own experiments, and it is why our constants are cross-validated on the target recordings instead.

5.3. Where the forward assumption fails

Eq. 2 assumes the performance moves forward, which is why the decision is tractable and also why a repeat or a jump breaks it. Following the protocol of [12] we splice three discontinuities into each performance at random destinations, re-running perception on the spliced audio so its conditioning is gen-

Table 3. Recovery from 41 spliced discontinuities. *rec@1*: fraction first tracked correctly within 1 s. *post*: onsets tracked within 1 s over the following 5 s.

Decision rule	break present		no break	
	rec@1	post	rec@1	post
argmax (1)	0.32	0.22	0.27	0.22
CANDOR	0.12	0.15	0.10	0.15
+ silence trigger	0.27	0.19	0.10	0.15
+ confidence trigger	0.15	0.22	0.20	0.23

uinely wrong afterwards, and excluding jumps across a page boundary where state resets anyway, leaving 41.

Table 3 reports a failure. After a jump CANDOR is worse than the rule it replaces, 0.15 against 0.22 of onsets tracked over the next five seconds, because the transition model holds the trajectory at a stale position. Suppressing it on detected silence helps where a break exists and not otherwise; triggering on the scorer’s own confidence needs no break and works in both cases, but only returns to argmax. A transition model worth eleven points on continuous playing is strong enough to be wrong at a discontinuity, and reconciling the two is open.

6. CONCLUSION

The synthetic-to-real gap in sheet-image score following has been read as a perception failure. It is a decision failure: the correct position is already among the candidates at every onset, and deciding sequentially among them takes real piano recordings from 80.0% to 94.4% of onsets within 0.5 s, against 88.3% for the previous best. Limitations: the page is given, the music is piano, and 16 pieces cannot resolve a point or two.

7. COMPLIANCE WITH ETHICAL STANDARDS

This is a numerical study that uses only publicly available data; no ethical approval was required.

8. ACKNOWLEDGMENTS

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